

It follows that

$$\begin{aligned} 1.4 &< \sqrt{2} < 1.5 \\ 1.41 &< \sqrt{2} < 1.42 \\ 1.414 &< \sqrt{2} < 1.415 \\ 1.4142 &< \sqrt{2} < 1.4143 \\ &\vdots \end{aligned}$$

These are sequences of rational numbers to the left and to the right of $\sqrt{2}$, both of which tend to $\sqrt{2}$, but never reach it. So $\sqrt{2}$ fills a certain gap. Indeed, with the irrational numbers added, we get a continuous line of increasing numbers, called the *real number line*. This property of having a *continuous* line is called *completeness* and we say that the real number line is **complete** (intuitively, it means that there are no gaps). This is in fact an axiom (called The Completeness Axiom), discussed later in the chapter.

Actually, there are *many* irrationals. It turns out that there are infinitely many rationals and infinitely many irrationals. They are all intertwined:

- between any two rationals $r_1 < r_2$ there are infinitely many irrationals,
- between any two irrationals $y_1 < y_2$ there are infinitely many rationals.

However, there are *more* irrationals than there are rationals. If we write the set of real numbers (the real number line) as the (disjoint) union of the rationals and the irrationals

$$\mathbb{R} = \mathbb{Q} \sqcup (\mathbb{R} \setminus \mathbb{Q})$$

then both $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ are infinite, but $\mathbb{R} \setminus \mathbb{Q}$ is ‘bigger’ in a certain sense. This statement is made rigorous using a mathematical theory called *measure theory*. It can be understood intuitively as follows: if a number is chosen randomly between, say, 0 and 1, then it is *almost surely* an irrational number (in other words, the probability of choosing a rational number is zero). So within the real number line there are ‘more’ irrationals. However there are also rationals *everywhere*. We say that **the rational numbers $\mathbb{Q}$ are dense within the reals $\mathbb{R}$.**

The ordering of real numbers

The real numbers are a **totally ordered set**: $\forall x, y \in \mathbb{R}$, one (and *only* one) of the following properties holds:

$$x = y \quad \text{or} \quad x < y \quad \text{or} \quad x > y.$$

We often use the following symbols for these important subsets of $\mathbb{R}$:

$$\begin{aligned} \mathbb{R}_+ &= \{x \in \mathbb{R} \mid x > 0\} \\ \mathbb{R}_- &= \{x \in \mathbb{R} \mid x < 0\} \end{aligned}$$

Infinity. It is convenient to introduce a symbol to help us express the fact that there is no greatest number. We thus introduce the symbol for **plus infinity**

$$+\infty,$$

which is thought of as an object that is greater than any real number. Similarly, **minus infinity**

$$-\infty$$

symbolizes an object that is smaller than any real number. Note that these are *not* numbers.

Intervals. The notion of an *interval* – a part of the real number line – will be very important:

Intervals

Let $a, b \in \mathbb{R}$ with $a \leq b$. Then the **closed interval** between a and b is defined as

$$[a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}.$$

Let $a, b \in \mathbb{R}$ with $a < b$. Then the **open interval** between a and b is defined as

$$(a, b) = \{x \in \mathbb{R} \mid a < x < b\}.$$

That is, closed intervals include all points between a and b , including a and b themselves. Open intervals do not include the endpoints. The points between a and b (excluding a and b themselves) are called **interior points**.

We can also define intervals that are closed on one end and open on the other. Let $a < b$, then

$$\begin{aligned} [a, b) &= \{x \in \mathbb{R} \mid a \leq x < b\}, \\ (a, b] &= \{x \in \mathbb{R} \mid a < x \leq b\}. \end{aligned}$$

Other important sets are **half-lines**, which are sets that have a lower/upper limit only on one side. Here the symbols for plus or minus infinity come in handy:

$$\begin{aligned} [a, +\infty) &= \{x \in \mathbb{R} \mid a \leq x\}, \\ (a, +\infty) &= \{x \in \mathbb{R} \mid a < x\}, \\ (-\infty, b] &= \{x \in \mathbb{R} \mid x \leq b\}, \\ (-\infty, b) &= \{x \in \mathbb{R} \mid x < b\}. \end{aligned}$$

The entire real line is often represented as the set of all points that are greater than $-\infty$ and less than $+\infty$:

$$\mathbb{R} = (-\infty, +\infty).$$

A natural question is the converse: which subsets of $\mathbb{R}$ deserve to be called an interval? The following lemma gives a simple characterization, and will be useful later (for instance, to show that the continuous image of an interval is itself an interval).

Lemma 2.5: Let $I \subseteq \mathbb{R}$ be a subset of $\mathbb{R}$ that satisfies the following condition: for any $x_1, x_2 \in I$ with $x_1 < x_2$, the entire closed interval $[x_1, x_2]$ is a subset of I . Then I is an interval, i.e. I could be an open, closed, or half-open-half-closed interval, and one or both of its endpoints can be infinite.

Proof. This is a simple proof (by contradiction) which we skip here. $\square$

Absolute value. An important operation that we will frequently encounter is the **absolute value** of a real number x :

$$|x| = \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0. \end{cases}$$

That is, $|x|$ measures the distance of x from 0, regardless of the sign of x . Consequently, we can define the **distance** between $x, y \in \mathbb{R}$ as:

$$|x - y| = \begin{cases} x - y & \text{if } x - y \geq 0 \text{ (i.e., if } x \geq y), \\ -(x - y) = y - x & \text{if } x - y < 0 \text{ (i.e., if } x < y). \end{cases}$$

Bounded sets of real numbers

The notion of a bounded set is an extension of the notion of an interval:

Bounded sets

- A subset $A \subseteq \mathbb{R}$ is said to be **bounded from above** if $\exists b \in \mathbb{R}$ such that

$$x \leq b, \quad \text{for all } x \in A.$$

Such a real number b is called an **upper bound** for A .

- $A \subseteq \mathbb{R}$ is said to be **bounded from below** if $\exists a \in \mathbb{R}$ such that

$$x \geq a, \quad \text{for all } x \in A.$$

Such a real number a is called a **lower bound** for A .

- A is called **bounded** if it is bounded from below and from above.
- If A is not bounded from above and not bounded from below, we say that A is **unbounded**.

Lower and upper bounds are not unique

It is very important to observe that both lower and upper bounds are not unique: if a is a lower bound for a subset $A \subseteq \mathbb{R}$, then any $y \leq a$ is also a lower bound. Similarly, if b is an upper bound for A , then any $z \geq b$ is also an upper bound.

Example 2.4: 1. The subset $A_1 = \{-2, 0.5, 7\}$ is bounded. Any number $a \leq -2$ is a lower bound and any number $b \geq 7$ is an upper bound.

2. The subset $A_2 = \mathbb{N}$ is bounded from below but not from above. Any number $a \leq 0$ is a lower bound.

3. The subset $A_3 = \mathbb{Z}$ is unbounded.

4. The subset $A_4 = \{\frac{1}{n} \mid n \in \mathbb{N}_+\} = \{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\}$ is bounded. Why? We see that a possible upper bound is 1, which belongs to A_4 . What about a lower bound? The number 0 is a lower bound, however 0 does not belong to A_4 . Below we will see that there is no lower bound that belongs to A_4 .

5. The subset $A_5 = \{x \mid |x| > 100\}$ is unbounded.

Let us take a closer look at the lower bound of $A_4 = \{\frac{1}{n} \mid n \in \mathbb{N}_+\}$:

Lemma 2.6: Any lower bound of the set $A_4 = \{\frac{1}{n} \mid n \in \mathbb{N}_+\}$ does not belong to A_4 .

Proof. By contradiction, suppose that $\exists a \in A_4$ such that $a \leq x$ for all $x \in A_4$. Since $a \in A_4$, there exists $N \in \mathbb{N}$ such that $a = \frac{1}{N}$. Observe that $\frac{1}{N+1}$ is also an element of A_4 and $\frac{1}{N+1} < \frac{1}{N} = a$. But this contradicts our assumption that a is a lower bound for A_4 . Hence no element in A_4 can be a lower bound of A_4 . $\square$

In fact, we can prove a stronger statement:

Lemma 2.7: The set $A_4 = \{\frac{1}{n} \mid n \in \mathbb{N}_+\}$ does not have a positive lower bound.

Proof. By contradiction, assume that there exists $r > 0$ that is a lower bound for A_4 . Define $N = \lceil \frac{1}{r} \rceil$ to be the first integer greater than or equal to $\frac{1}{r}$. Then $N + 1 > \frac{1}{r}$ and consequently $\frac{1}{N+1} < r$. But $\frac{1}{N+1} \in A_4$, in contradiction to the assumption that r is a lower bound. Hence there is no positive lower bound, and 0 is the *greatest lower bound*. $\square$

Supremum and infimum

Let $A \subset \mathbb{R}$ be a subset.

- The **supremum** (if it exists) of A (also called the **least upper bound**, **l.u.b.**) is the smallest of all upper bounds of A . It is denoted

$$s = \sup A$$

and it fulfils the following two conditions:

1. $\forall x \in A, x \leq s$,
2. $\forall r < s \exists x \in A \text{ s.t. } x > r$.

If there is no such number, we define $\sup A = +\infty$.

- The **infimum** (if it exists) of A (also called the **greatest lower bound**, **g.l.b.**) is the largest of all lower bounds of A . It is denoted

$$\ell = \inf A$$

and it fulfils the following two conditions:

1. $\forall x \in A, x \geq \ell$,
2. $\forall r > \ell \exists x \in A \text{ s.t. } x < r$.

If there is no such number, we define $\inf A = -\infty$.

Note that in the above definition, we defined *the* supremum and *the* infimum, implying that these two numbers are unique. *A priori*, that is not obvious (even though it is true). It requires a proof.

Proposition 2.8: For any subset $A \subseteq \mathbb{R}$, there are unique elements $\ell, s \in \{-\infty\} \cup \mathbb{R} \cup \{+\infty\}$ such that $\ell = \inf A$ and $s = \sup A$.

Proof. The *existence* of these elements is guaranteed by the *Completeness Axiom*, stated immediately below. The *uniqueness* of these elements is left as an exercise. *Hint: prove it by contradiction.* $\square$

The Completeness Axiom

The property described above — that the real number line has “no gaps” — is in fact an *axiom*: it cannot be derived from the other properties of $\mathbb{R}$ (it is precisely the property that $\mathbb{R}$ has and $\mathbb{Q}$ does not). We state it here in the form we will use throughout these notes:

Completeness Axiom. Every nonempty subset $A \subseteq \mathbb{R}$ that is bounded above (resp. below) has a supremum $\sup A \in \mathbb{R}$ (resp. infimum $\inf A \in \mathbb{R}$).